Project Initiation: Document Tampering

1. Team Details

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2. Real-Life Problem Statement

The increasing use of digital documents in banking, education, healthcare, and other sectors has also increased the risk of document forgery and unauthorized modifications. Changes to text, numbers, dates, or specific regions of a document can be difficult to detect through manual inspection.

This project aims to develop an automated document forgery and tampering detection system using image preprocessing, ELA, CNN, OCR, and text-based analysis. The system will analyze both visual and textual features to classify a document as authentic or tampered, while also supporting tampered-region identification and model explainability using ground-truth masks and techniques such as Grad-CAM.

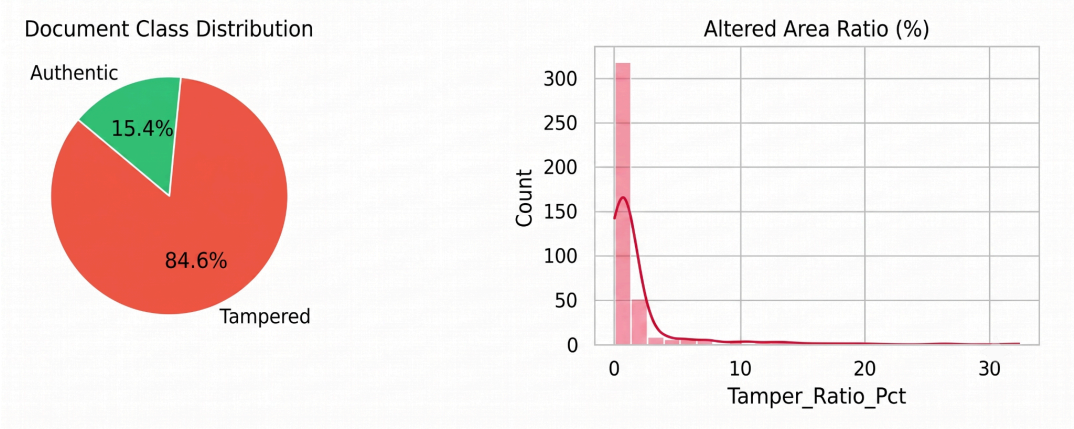
3. Completed Dataset [DocTamper]

- **Source:** DocTamper Dataset (derived from research on large-scale document image tampering detection and hosted via academic repository/Kaggle mirrors).
- **Composition & Volume:** A curated subset of 1,20,000 scanned documents (receipts, financial statements, bills, and ID forms) formatted at uniform spatial resolutions (512 x 512). Binary ground-truth tampering masks for every image, where pixel value 1 represents forged/tampered regions and 0 represents authentic regions. Features an inherent real-world class distribution where the majority of documents are tampered due to multiple synthetic variations generated per source, alongside authentic documents containing splicing, text erasure, or copy-move manipulations.
- **Data Access & Storage:** The raw dataset and processed subsets are compiled locally into structured directories (/data/train, /data/val, /data/test) and mirrored on a private repository storage bucket for reproducible batch loading.

4. Exploratory Data Analysis (EDA)
Based on a randomized sampling of 500 document transactions from the total 120,000 records in the DocTamper LMDB database, we conducted a full Exploratory Data Analysis.

A. Data Distributions

- **Document-Level Class Imbalance:** The random sampling revealed a severe class imbalance within the dataset. As shown in the Class Distribution chart, 84.6% of the sampled documents are tampered, while only 15.4% are authentic. This indicates that the dataset relies heavily on multiple synthetic manipulations per authentic source document. To prevent the model from simply predicting the majority class, our training pipeline must implement class-weighted loss or oversample the authentic documents.
- **Spatial Class Imbalance:** The "Altered Area Ratio" histogram demonstrates that manipulated pixels occupy a highly sparse distribution. The vast majority of forgeries alter less than 10% of the total document surface area. This extreme pixel-level imbalance means standard classification loss will fail for segmentation tasks; the model will require region-weighted spatial loss functions to penalize background dominance.



B. Missing Values & Data Integrity

- **Integrity Check:** Automated byte validation over the LMDB binary transactions yielded exceptional data integrity. Out of the 500 random samples, the pipeline detected 0 missing labels and 0 corrupted images.
- **Leakage Control:** Only 1 duplicate file was detected (Duplicates: 1), confirming that the dataset is highly unique and ready for tensor conversion without the need for complex data imputation.

C. Preprocessing

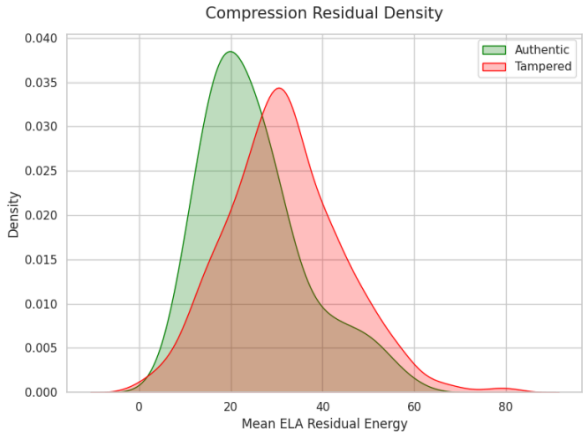
To expose digital manipulations not visible to the naked eye, we engineered a multi-modal preprocessing pipeline:

- **Error Level Analysis (ELA):** Scans are re-saved at a 90% JPEG quality threshold and subtracted from their original pixel matrices to amplify compression artifacts (using a 25x multiplier).
- **Laplacian Edge Filtering:** A high-pass convolution kernel is applied to extract high-frequency micro-edge noise across digital splicing boundaries.
- **Ground-Truth Binarization:** The provided label arrays are converted into strict binary masks (using a > 127 pixel intensity threshold) to serve as targets for pixel-segmentation loss calculations.

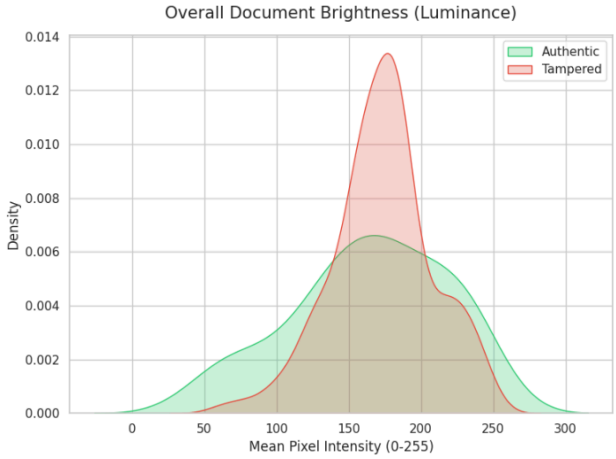


D. Feature Insights & Separability

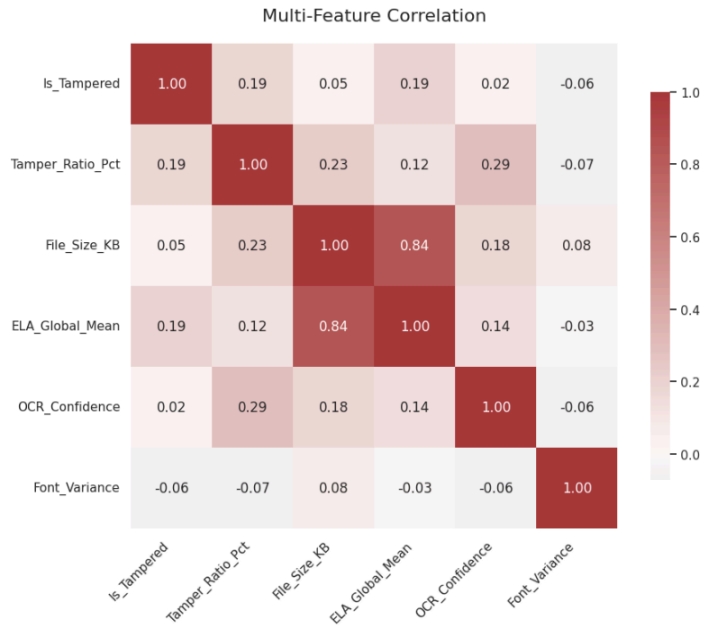
- **Compression Residual Separability:** The Compression Residual Density plot proves distinct mathematical separability between authentic and tampered zones. Pristine document regions (green) exhibit a tight, low-variance distribution of ELA residuals. In contrast, tampered pages (red) exhibit a much broader distribution shifted toward higher residual energy, confirming ELA as a highly discriminative input channel.



- **Luminance Distribution:** Analysis of pixel luminance reveals nearly identical, overlapping density curves for both authentic and tampered documents. This confirms forgeries lack a distinct global brightness anomaly and blend seamlessly into the document's color space. Consequently, simple visual thresholding is insufficient for detection, directly justifying our need for advanced forensic preprocessing (like ELA) and Convolutional Neural Networks to expose hidden, localized artifacts.



- **Feature Correlation:** The Multi-Feature Correlation heatmap reveals critical relationships for model building:
    - **File Size & ELA Residual Energy:** File_Size_KB exhibits a very strong positive correlation (**0.84**) with ELA_Global_Mean, indicating that higher-resolution files naturally retain higher baseline compression residual energies.
    - **Tampering & Altered Area:** The Is_Tampered flag shows a positive correlation (**0.19**) with Tamper_Ratio_Pct, reflecting the statistical relationship between binary document alteration and total surface area coverage.
    - **Tampering & Compression Residuals:** Is_Tampered also shows a positive correlation (**0.19**) with ELA_Global_Mean, verifying that digital splicing introduces consistent compression anomalies compared to pristine pages.
    - **Tampered Area & OCR Confidence:** Tamper_Ratio_Pct shows a positive correlation (**0.29**) with OCR_Confidence, indicating that larger structural regions modified on the document surface tend to trigger distinct text extraction patterns.



5. Probable Methodology

The system will examine a document in both visual and textual ways to detect possible forgery or tampering.

- **Preprocessing:** Prepare and standardize the document images for analysis, including Laplacian edge filtering.

- **ELA:** Identify unusual image-compression patterns that may indicate editing.
- **CNN:** Learn visual features from the document using a U-Net architecture and classify it as authentic or tampered.
- **OCR:** Extract the text present in the document for further analysis.
- **Text Analysis:** Check the extracted text for patterns or inconsistencies related to tampering (e.g., OCR confidence drops, font height variance).
- **Feature Fusion:** Combine visual and text-based information to improve detection.
- **Final Classification:** Predict whether the document is authentic or tampered using convolutional neural networks and an autoencoder anomaly check.
- **Region Analysis:** Use the provided tampering masks to compare the predicted and actual tampered regions. Grad-CAM may also be used to understand which parts of the document influenced CNN's decision.

The primary deep learning technique planned for the project is CNN-based image analysis and classification.

6. Expected Outcome & Performance Targets

The main goal is to develop a system that can automatically identify whether a document is authentic or tampered with and highlight the regions that may have been modified.

Expected Outcomes:

- Develop an automated pipeline for document forgery detection.
- Classify documents as authentic or tampered using visual and textual features.
- Identify and visualize suspicious or tampered regions using ground-truth masks and Grad-CAM.
- Evaluate the model using standard classification and localization metrics.
- Study the limitations of OCR, image quality, and possible dataset bias.

Performance Targets:

- **Classification:** Target Macro F1-Score $\geq$ **0.90** and AUC-ROC $\geq$ **0.93**.
- **Tampered Region Detection:** Target IoU $\geq$ **0.65** and Dice Score $\geq$ **0.75** (specifically chosen to handle the extreme spatial class imbalance discovered during EDA).
- **Inference Time:** Aim for approximately 1.5 seconds or less per document page on standard hardware.